UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2010 question paper

for the guidance of teachers

9701 CHEMISTRY

9701/23 Paper 2 (AS Structured Questions), maximum raw mark 60

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1 **(a)** enthalpy change when 1 mol of a compound is formed (1)
from its elements (1)
in their standard states under standard conditions (1) [3]

(b) **(i)**

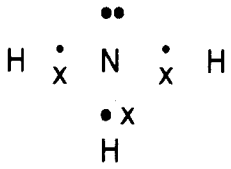
	$N_2H_4(l)$ +	$O_2(g)$ →	$N_2(g)$ +	$2H_2O(g)$
ΔH_f^o/kJ mol^{-1}	+50.6			–241.8

$\Delta H^o_{reaction}$ = 2(–241.8) – (+50.6) (1)
= –534.2 kJ mol^{-1} (1)

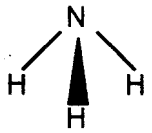
(ii) E_a is too high (1)

(iii) products are H_2O and N_2 which are harmless/non toxic
or are already present in the atmosphere (1) [4]

(c) **(i)** 'dot-and-cross' diagram (1)

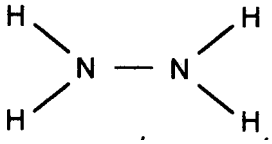


(ii)



(1)

(iii) minimum is



allow bond angle around N atom between 109° and 104° (1) [4]

(d) –2 (1) [1]

[Total: 12]

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2 **(a)** the energy required to remove one electron from each atom (1)
in one mole of gaseous atoms (1)
or
the enthalpy change in kJ mol^{-1} for (1)
$M(g) \rightarrow M^+(g) + e^-$ (1) [2]

(b) **(i)** first ionisation energy decreases down Group 1 (1)
outermost electron is further from nucleus
or has greater shielding (1)

(ii) outermost electron experiences less attraction
or formation of M^+ cation becomes easier down Group 1 (1) [3]

(c) **(i)** $n(Li) = \frac{0.83}{6.9} = 0.12$ (1)

(ii) 2 mol Li $\rightarrow$ 1 mol H_2
0.12 mol Li $\rightarrow \frac{1 \times 0.12}{2} = 0.06$ mol H_2 (1)
volume of $H_2 = 0.06 \times 24.0 = 1.44 dm^3$ (1)

(iii) 2 mol Li $\rightarrow$ 2 mol LiOH
0.12 mol Li $\rightarrow$ 0.12 mol LiOH in 0.50 dm^3 (1)
$[LiOH] = \frac{0.12 \times 1}{0.50} = 0.24$ mol dm^{-3} (1) [5]

(d) sodium burns with a yellow flame
or white solid formed
or colour of chlorine disappears (1)

$2Na + Cl_2 \rightarrow 2NaCl$ (1) [2]

[Total: 12]

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3 **(a)** **(i)** Ca (1)

(ii) S **or** C [allow H (H_2O_2) or N (NO, NO_2)] (1)

(iii) He (1)

(iv) A*l* (1)

(v) Si **or** Ge (1)

(vi) A*l* (1) [6]

(b) any **two** from N **or** O **or** F (1) [1]

(c) **(i)** Al_2O_3 **or** SiO_2 (1)

(ii) SO_2 **or** P_2O_3/P_4O_6 (1)
and and
SO_3 **or** P_2O_5/P_4O_{10} (1)

(iii) Na_2O (1)

(iv) Al_2O_3 (1) [5]

[Total: 12]

4 **(a)** **reaction 1** free radical substitution (1)

reaction 2 elimination (1) [2]

(b) **(i)** in reaction 4 $CH_3C(OH)(CN)CH_3$ (1)

(ii) in reaction 3 I^- (1)

(iii) in reaction 3 CH_3I
or in reaction 4 CH_3COCH_3 (1) [3]

(c) a species which has a lone pair of electrons
or which reacts with an electron deficient (δ+) centre in a molecule (1) [1]

(d) in reaction 3 OH^- (1)
in reaction 4 CN^- (1) [2]

(e) π bonding is electron rich (1) [1]

[Total: 9]

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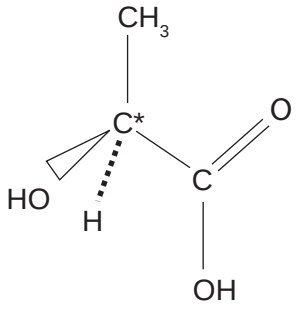
5 (a)



[3]

(b)

	reagent(s)	condition(s)
step 1	$Cr_2O_7^{2-}$ / H^+ (1)	distil off aldehyde (1)
step 2	HCN in presence of CN^- **or** KCN + dil H_2SO_4 (1)	room temperature (1)
step 3	aqueous mineral acid/ /H_2SO_4/HC/ **not** HNO_3 (1)	heat under reflux (1)

in **each** case, the reagent must be correct before the condition mark is awarded [6]

(c) (i) a protein (1)

(ii) 2,4-dinitrophenylhydrazine/Brady's reagent (1)
yellow-orange-red ppt. (1)

(iii) acidified $K_2Cr_2O_7$ **or** Lucas test **or** CH_3CO_2H/H^+ (1)
colour changes **or** cloudiness **or** fruity smell
from orange to green (1)

(iv) $LiA/H_4/NaBH_4$
or H_2/Ni etc. (1) [6]

[Total: 15]

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